SOMAIYA VIDYAVIHAR UNIVERSITY

23·05·2025 (M)

Semester: January 2025– April 2025			
Maximum Marks: 50	**Examination: End-Semester Examination (Re-ESE)**		**Duration: 2 Hrs.**
Programme code: 01 **Programme: BTech in Computer Engineering.**	**Class:**	**LY**	**Semester:** ~~I/II/III/IV/V/VI/VII/~~ VIII (SYU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department/Section/Center:** COMP		
Course Code: 116U01E864	**Name of the Course: Natural Language Processing**		
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	How bi-gram and trigram Language Model works for language understanding? Explain with example.	**05**
ii)	What is the role of Zipf's Law for handling stop words in preprocessing?	**05**
iii)	Describe the syntactic constraints on coreference ?	**05**
Q.2	Attempt any **one**	**10**
i)	How Finite state transducer works for morphological parsing? Explain derivational and inflectional morphology with suitable examples.	
ii)	Explain how the POS tagging problem can be modeled as a Hidden Markov Model (HMM). Discuss the representation of states, observations, transition probabilities, and emission probabilities for this model.	
Q.3	Attempt any **one**	**10**
i)	How the training process of skip gram model works in word embedding? What is the objective function optimized during training?	
ii)	How Lesk's algorithm works for Word Sense Disambuigation. How is it different than Walker's algorithm ? Justify with suitable example.	
Q.4	Attempt the following	**10**
	Consider the following PCFG fragment: S → NN VP 0.50 S → VP NN 0.50 NP → NN PB 0.40 PB → PP NN 0.30 VP → VB NN 0.30 VP → VB NP 0.20 VP → NN VB 0.25 VP → NN PB 0.15 PP → with 0.10 PP → without 0.10 VB → play 0.30 VB → enjoy 0.20 VB → watch 0.25 NN → children 0.15 NN → cricket 0.15 NN → friends 0.20 NN → football 0.10 NN → music 0.12	

	Using CKY algorithm, find the probability score for the most probable tree for the sentence S = **"children enjoy cricket with friends".**	
Q.5	Write short notes on the following. a) Sentiment analysis b) Text summarization along with different types.	10